



AI, GenAI & Agentic AI for Automotive Leaders and Developers

Surendra Panpaliya

Generative AI

Gen-AI



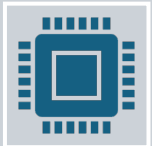
Prerequisites



Basic knowledge of Python (functions, control flow)



Familiarity with automotive workflows,



systems, and product lifecycles



Lab Setup Requirements



Laptop with Python 3.10+, Jupyter Notebook, VS Code



Libraries: numpy, pandas, scikit-learn, matplotlib, openai, langchain, chromadb



Access to Azure OpenAI or Google Gemini APIs



Internet connectivity, Excel, and sample automotive datasets

Trainer Deliverables



Pre-reading PDFs on AI, GenAI, Prompting



Setup guide for labs (for technical team)



Sample notebooks and visual examples



Hackathon guidance



Final review and open Q&A

Program Outcomes for Management Teams



Clearly understand AI, GenAI, and Agentic AI in business context



Evaluate where to invest in AI inside your automotive enterprise



Identify use cases with measurable ROI



Build leadership confidence in leveraging AI responsibly



Align technical and business teams on AI vision

Day-Wise Outline



Day 1: AI/ML/DL – Foundation for Automotive Innovation



Day 2: AI Leadership, Risk Management & Storytelling with Data



Day 3: Generative AI and Document Automation



Day 4: Strategic Thinking – Competing in the AI Age



Day 5: Introduction to Agentic AI and Innovation Hackathon

Day 3



Generative AI and Document Automation



Objective: Simplify GenAI and LLMs



For creating documents, manuals,



Customer communication.

Day 3

What is Generative AI (vs Traditional AI)

LLMs: GPT, Gemini, Claude

What leaders need to know

Prompt Engineering:

Asking the right questions to the AI

Day 3



Use Case: Auto-generate driver manuals



from technical specs



Demo: Prompt Gemini



to create onboarding documents



Case Study: Using GenAI



for multilingual manual generation





The market size of Generative AI in the automotive industry is projected to reach around USD **2,691.92 million** by 2032, with a CAGR of 24.03%.

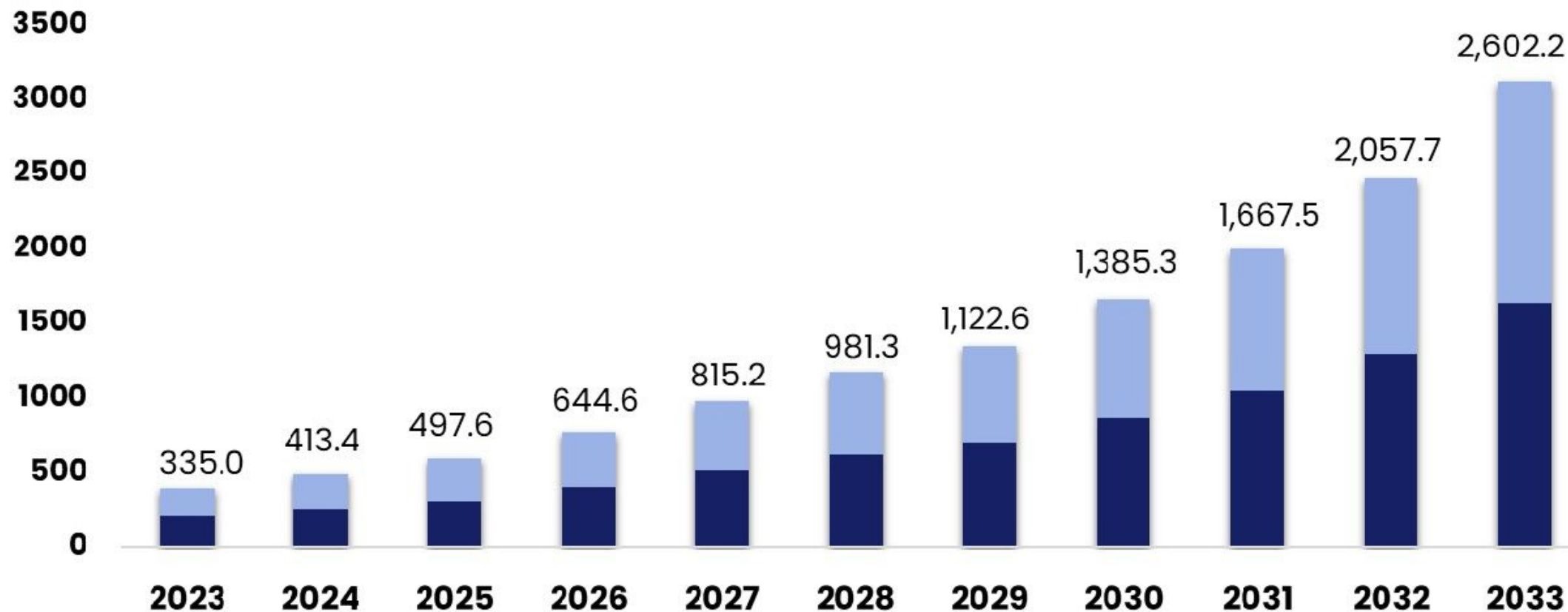


Generative Ai In Automotive Market

Size Based on Deployment Mode, 2023-2033 (USD Million)

■ On-Premise

■ Cloud-Based



The Market will Grow
At the CAGR of:

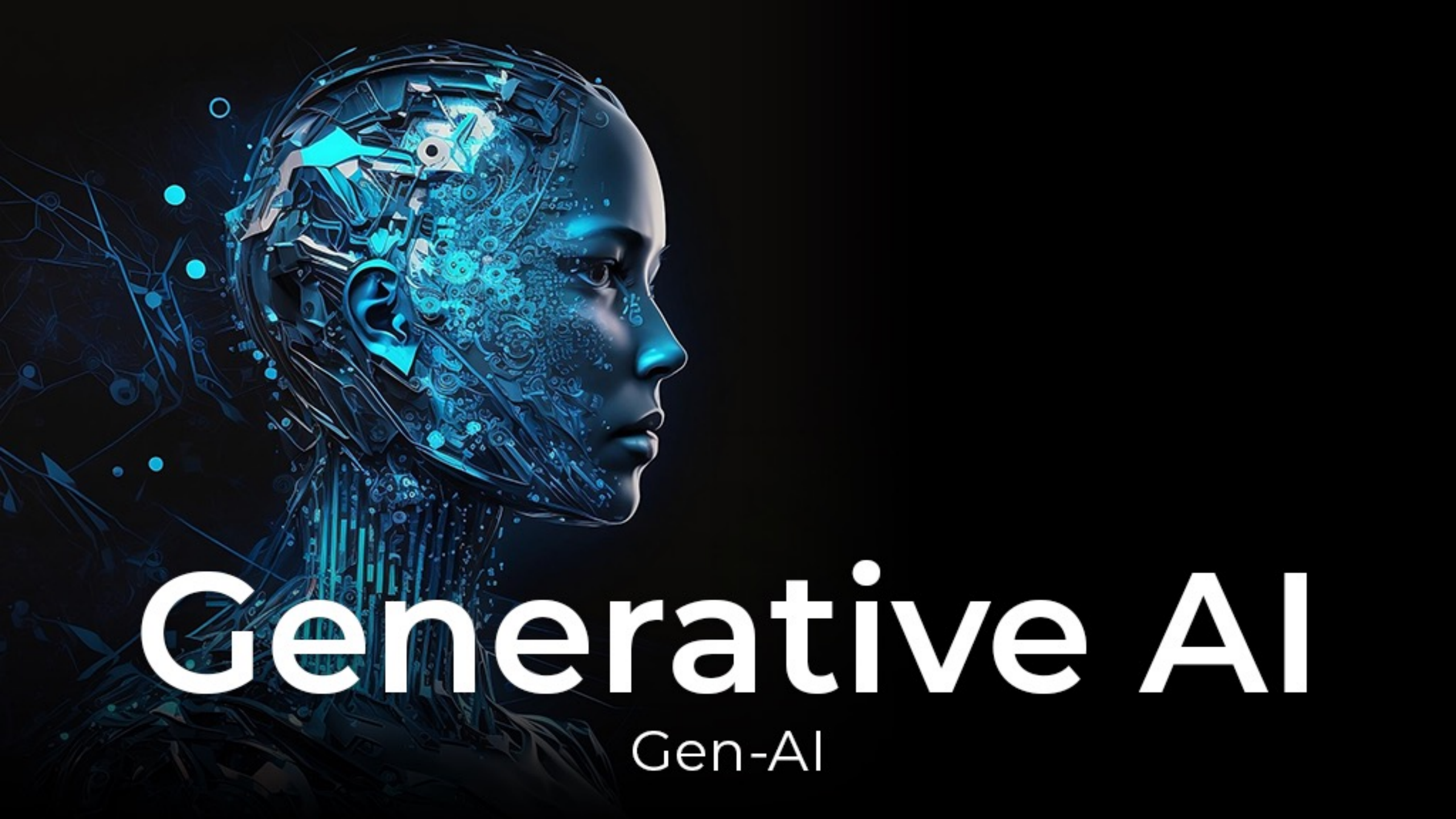
23.4%

The forecasted market
size for 2033 in USD

\$2,602.2M



MarketResearch
WIDE RANGE OF GLOBAL MARKET



Generative AI

Gen-AI

What is Generative AI?



CLASS OF ARTIFICIAL
INTELLIGENCE MODELS



CAN **GENERATE NEW
CONTENT**



SUCH AS TEXT, IMAGES,
AUDIO, VIDEO, OR CODE



BY LEARNING PATTERNS
FROM EXISTING DATA.

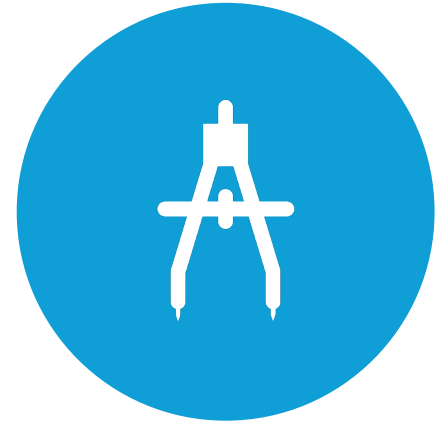
What is Generative AI?



MODELS DON'T JUST
ANALYZE DATA



THEY **CREATE**
SOMETHING NEW



MEANINGFUL FROM IT.

What is Generative AI?

At the core of GenAI are **foundation models**

GPT (for text),

DALL·E or **Stable Diffusion** (for images),

MusicLM (for audio),

Codex (for code generation), and more.

What is Generative AI?



MODELS USE DEEP
LEARNING,



ESPECIALLY
TRANSFORMERS,



TO UNDERSTAND AND
GENERATE CONTENT



THAT MIMICS HUMAN
INTELLIGENCE.

Generative AI vs Traditional AI

Feature	Traditional AI	Generative AI
Goal	Make predictions or classifications	Generate new content: text, images, code, etc.
Input → Output	Input → Fixed output (e.g., yes/no, number)	Input → Creative output (e.g., paragraph, summary)
Training	Rule-based or statistical	Trained on vast internet-scale datasets
Example Tool	ML model to predict brake failure	LLM (e.g., GPT-4) to generate a troubleshooting guide

Simplified Example (Automotive)

Traditional AI

Predict if a brake system will fail

based on sensor data

Output: Yes / No / Probability

Generative in AI Automotive



Automatically generate a technical document



What to do if your EV's regenerative



braking system fails



Output:



A full, human-like document tailored to the issue

Generative AI in the **Automotive Industry**





8 Impactful Use Cases of Generative AI in the Automotive Industry



Generative
AI-Powered Chatbot



Efficient
Roadside Assistance



In-Car Virtual
and Voice Assistants



Predictive
Maintenance

Use Cases of Generative AI for Automotive



Car Design
and Prototyping



Manufacturing
Process Enhancement



Supply Chain
Optimization



Virtual Testing
and Simulation



1. Document Automation

Task	GenAI Output
Create a service manual	“How to replace the battery in Model X EV”
Draft technical SOPs	“Engine overheating SOP for workshop teams”
Auto-fill technical compliance docs	Generate ISO/OBD/ADAS checklists
Translate documents in 10+ languages	Localized manuals in Hindi, German, Japanese



1. Document Automation



TOOLS



OPENAI GPT,



CLAUDE,



GEMINI,



MICROSOFT
COPILOT



2. Customer Communication

Task	GenAI Output
Auto-generate replies to FAQs	“What is covered under my EV warranty?”
Personalized service reminders	“Dear Mr. Raj, your Model Y service is due...”
Chatbots for troubleshooting	Real-time AI-powered help with navigation or charging



2. Customer Communication



Tools:



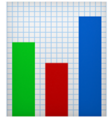
ChatGPT API,



Azure OpenAI
Service,



Rasa with LLMs



3. Marketing & Sales Enablement

Task	GenAI Output
Create a product brochure	“Top features of the new hybrid SUV”
Summarize reviews	“Customer sentiment on the ADAS system”
Compare models dynamically	“Model A vs Model B: Which EV suits your needs?”



4. Factory Floor Enablement

Task	GenAI Output
Generate real-time operator checklists	“Daily 9-step inspection for Battery Line 3”
Voice-to-text reports	Verbal dictation into formatted maintenance logs
Quick SOP generation from audio/text	Auto-create a standard procedure from notes

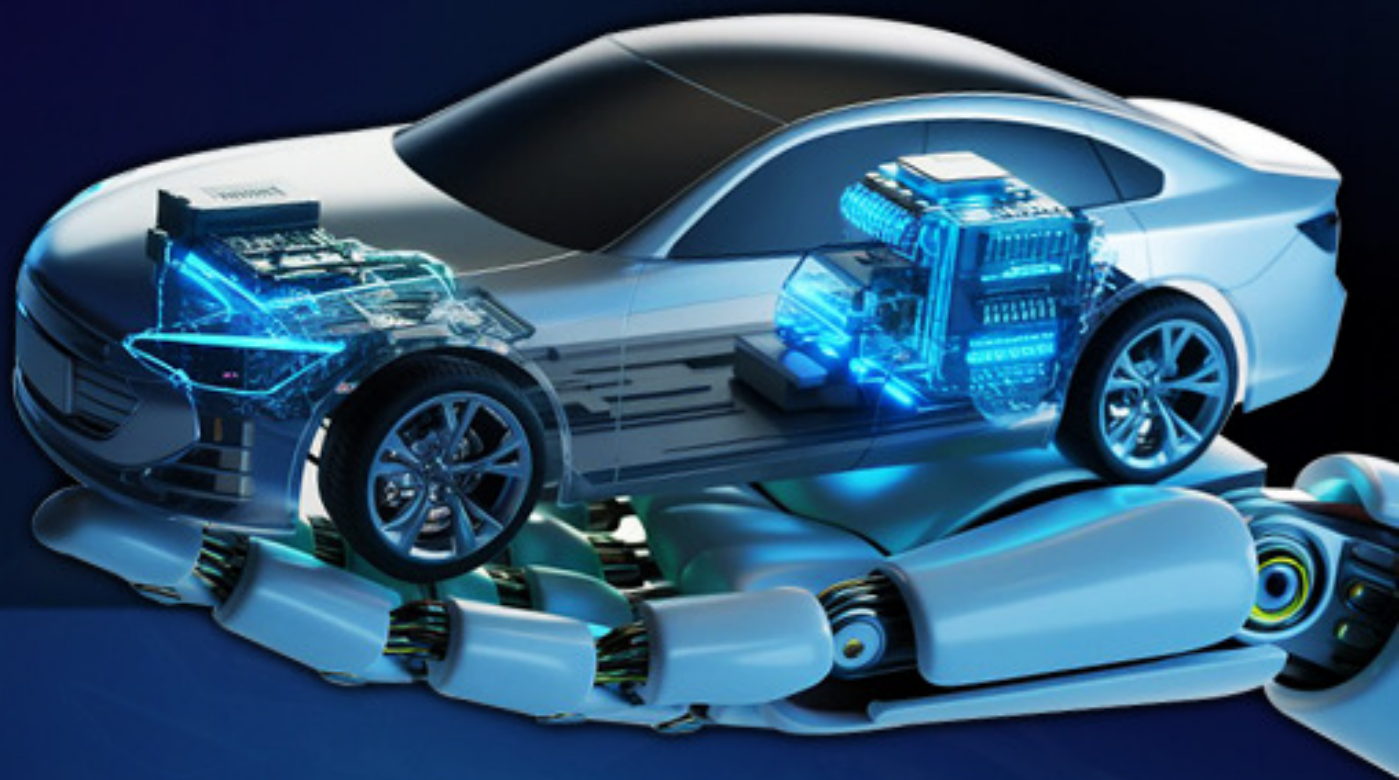
Benefits of GenAI for Automotive Industry

Benefit	Impact
 Speed	Documents/manuals created in minutes, not weeks
 Multilingual	Instantly localize communication globally
 Cost	Reduces dependency on manual writers/translators
 Consistency	Unified tone, format, and compliance across teams
 Intelligent Automation	Pulls from real-time data to create live documentation



ROLE OF

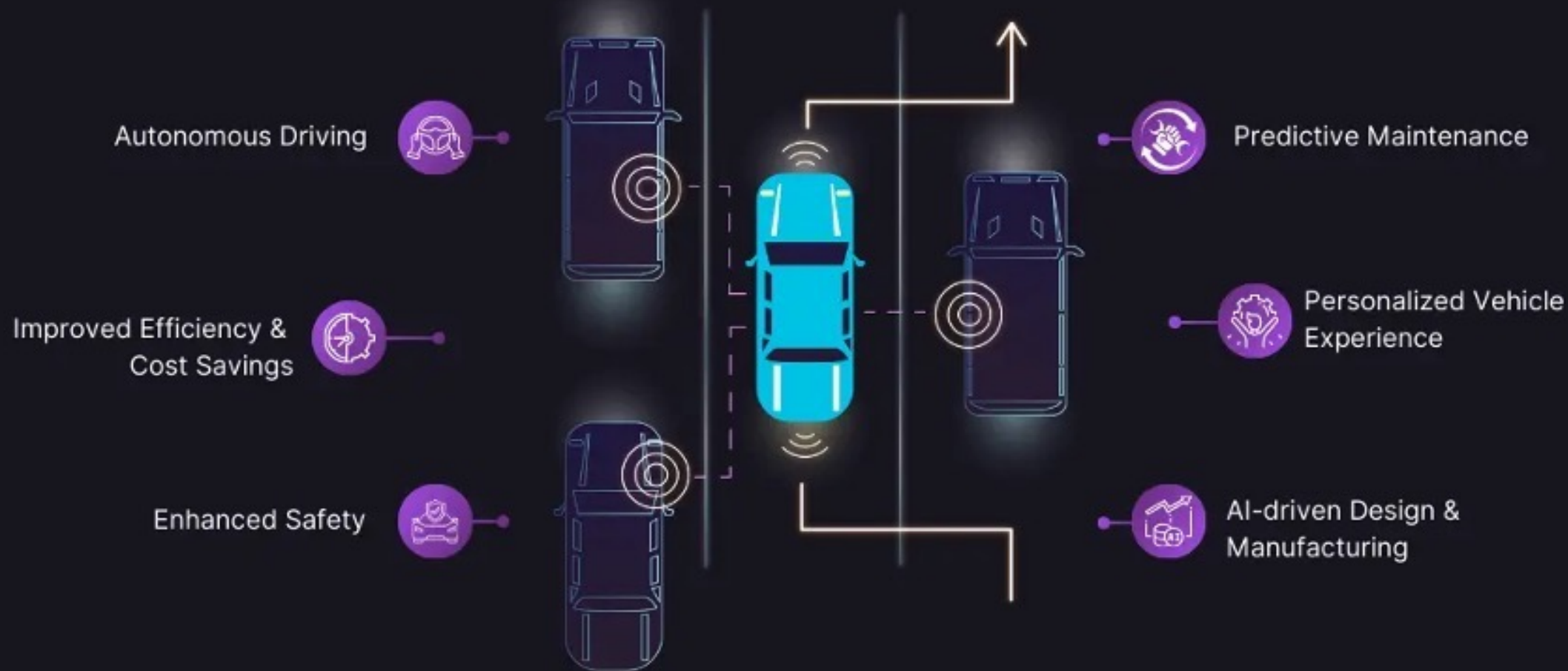
Generative **AI** in the **Automotive** Industry



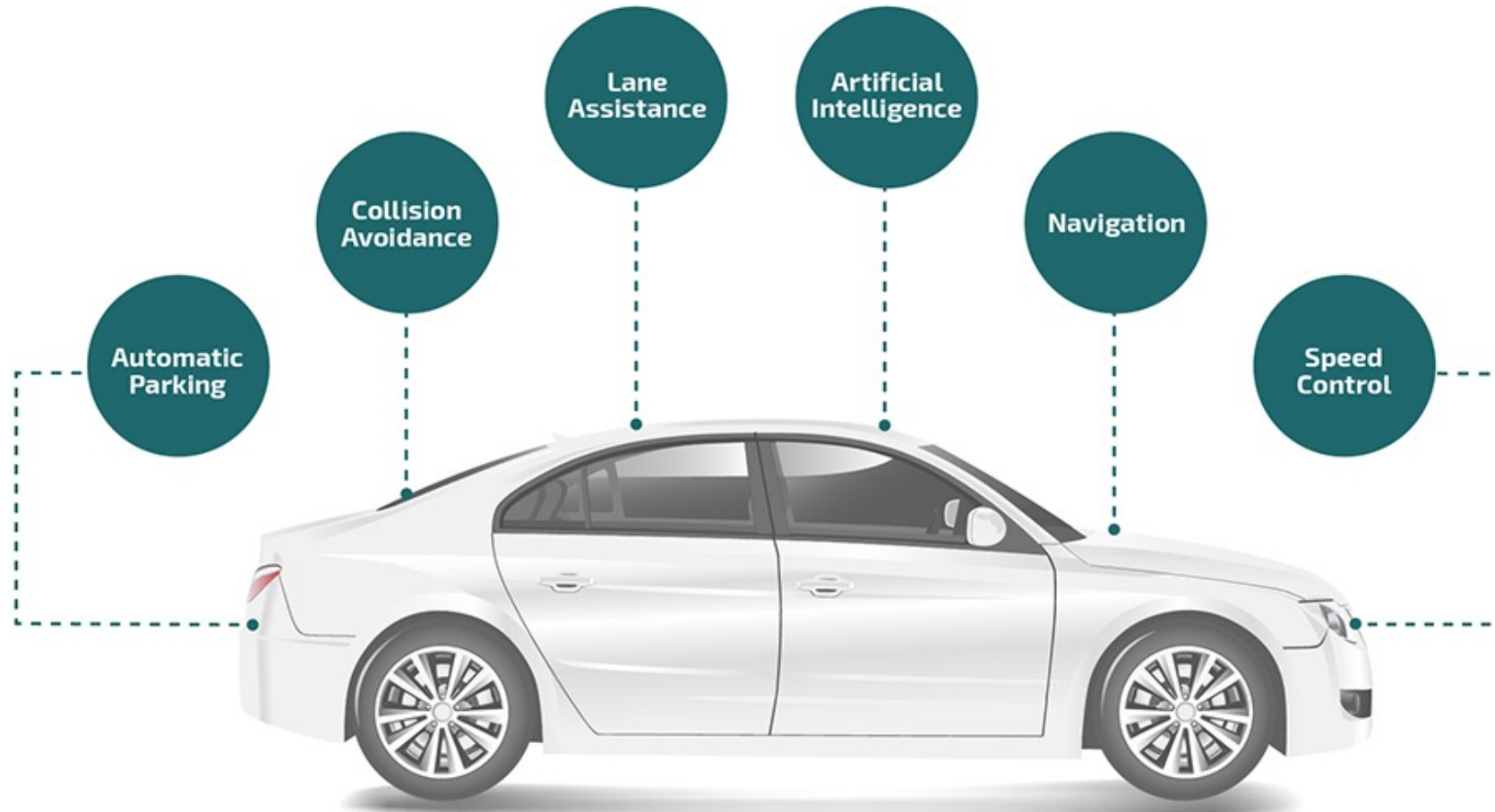
GENERATIVE AI USE CASES IN THE **AUTOMOTIVE INDUSTRY**



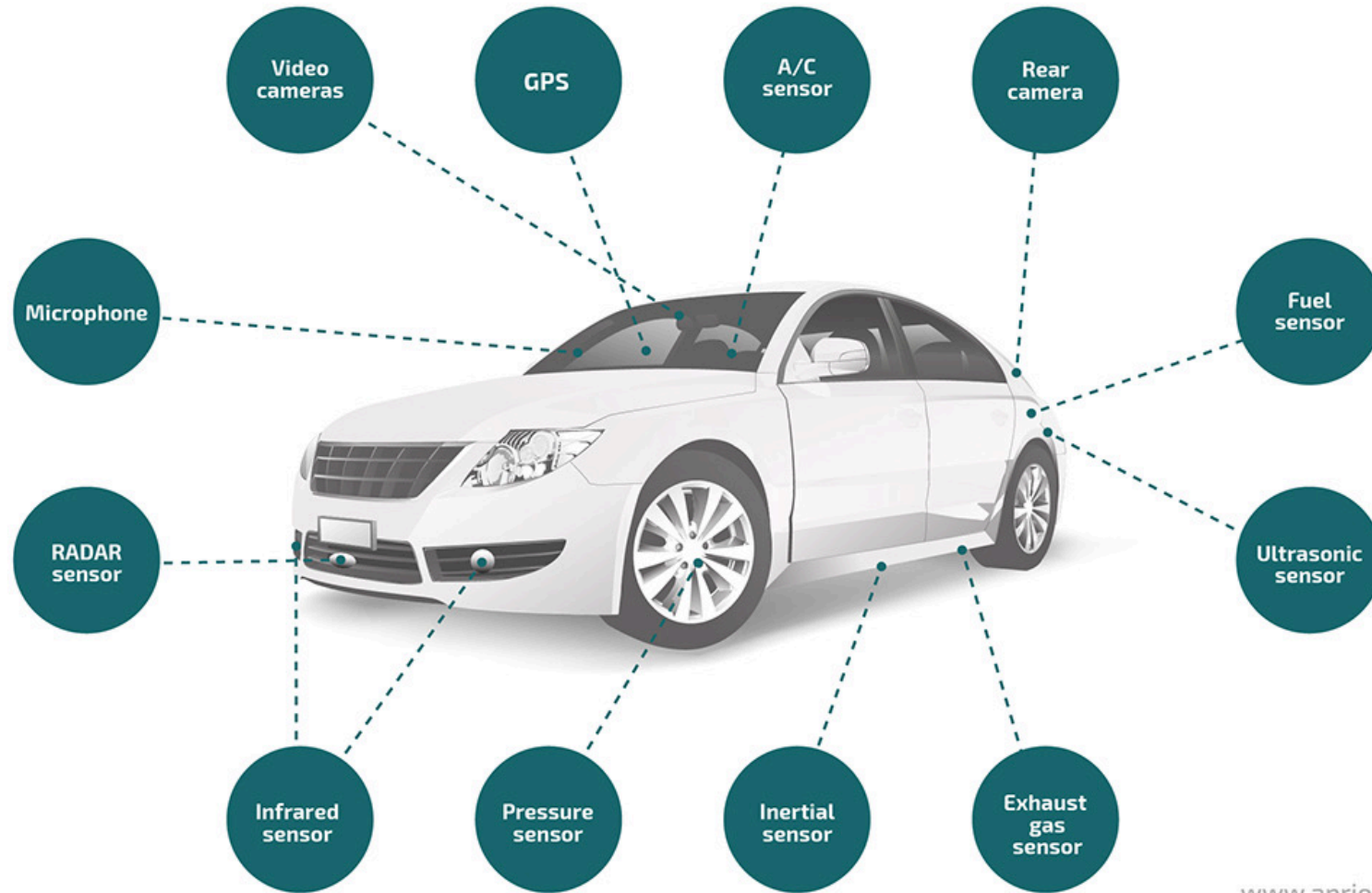
Why Should Businesses Adopt **AI in Automotive**?



6 COMMON ADAS TASKS



MOST COMMON TYPES OF IN-VEHICLE SENSORS



Real-World Example



Company: Toyota



Use Case: Automating service guides for 50+ hybrid variants



Before: Each guide was manually written in 5 languages

Real-World Example

After GenAI:

One master prompt

Translated to 10 languages

80% reduction in time and manual effort

Integrated with service chatbot for technicians

Future Vision: GenAI in the Car



Ask your vehicle: How do I check tire pressure?



Vehicle's in-built GenAI generates an



interactive answer with images



from your specific model.

Future Vision: GenAI in the Car



Ask your AI manual



“Show me how to replace the cabin air filter.”



AR/VR + GenAI generates a



Step-by-step animated guide.

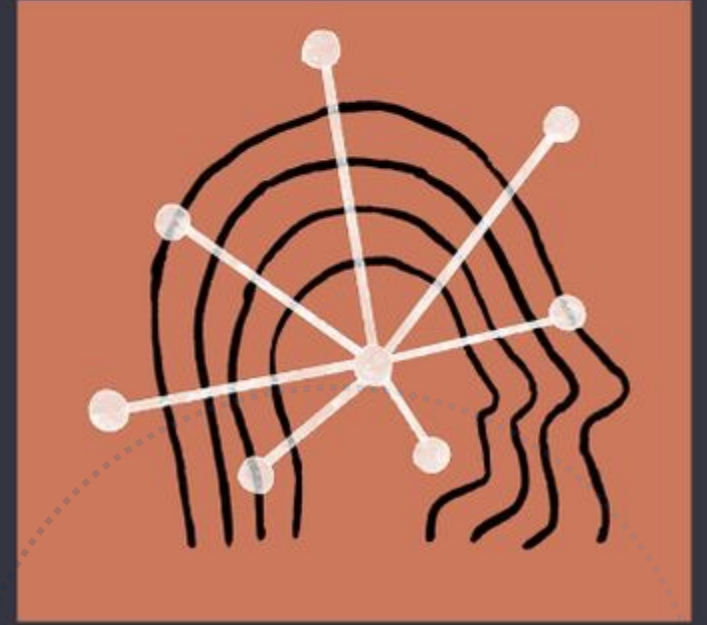
LLMs: GPT, Gemini, Claude



What leaders need to know?



Gemini

The Gemini logo is displayed on a black rectangular background. The word "Gemini" is written in a sans-serif font, with the "G" in blue and the remaining letters in white. A small white four-pointed star is positioned above the "i".

Gemini vs ChatGPT vs Claude

LLMs: GPT, Gemini, Claude



Objective:



Help decision-makers understand how to



leverage LLM like GPT (OpenAI),



Gemini (Google), and Claude (Anthropic)



to drive innovation, efficiency, and competitive edge in automotive.

What Are LLMs?



AI SYSTEMS TRAINED



ON MASSIVE
AMOUNTS OF DATA



BOOKS, MANUALS,
WEB CONTENT, CODE

What Are LLMs? (In Simple Words)



UNDERSTAND
NATURAL
LANGUAGE



ANSWER
QUESTIONS



GENERATE TEXT,
CODE,
DOCUMENTATION



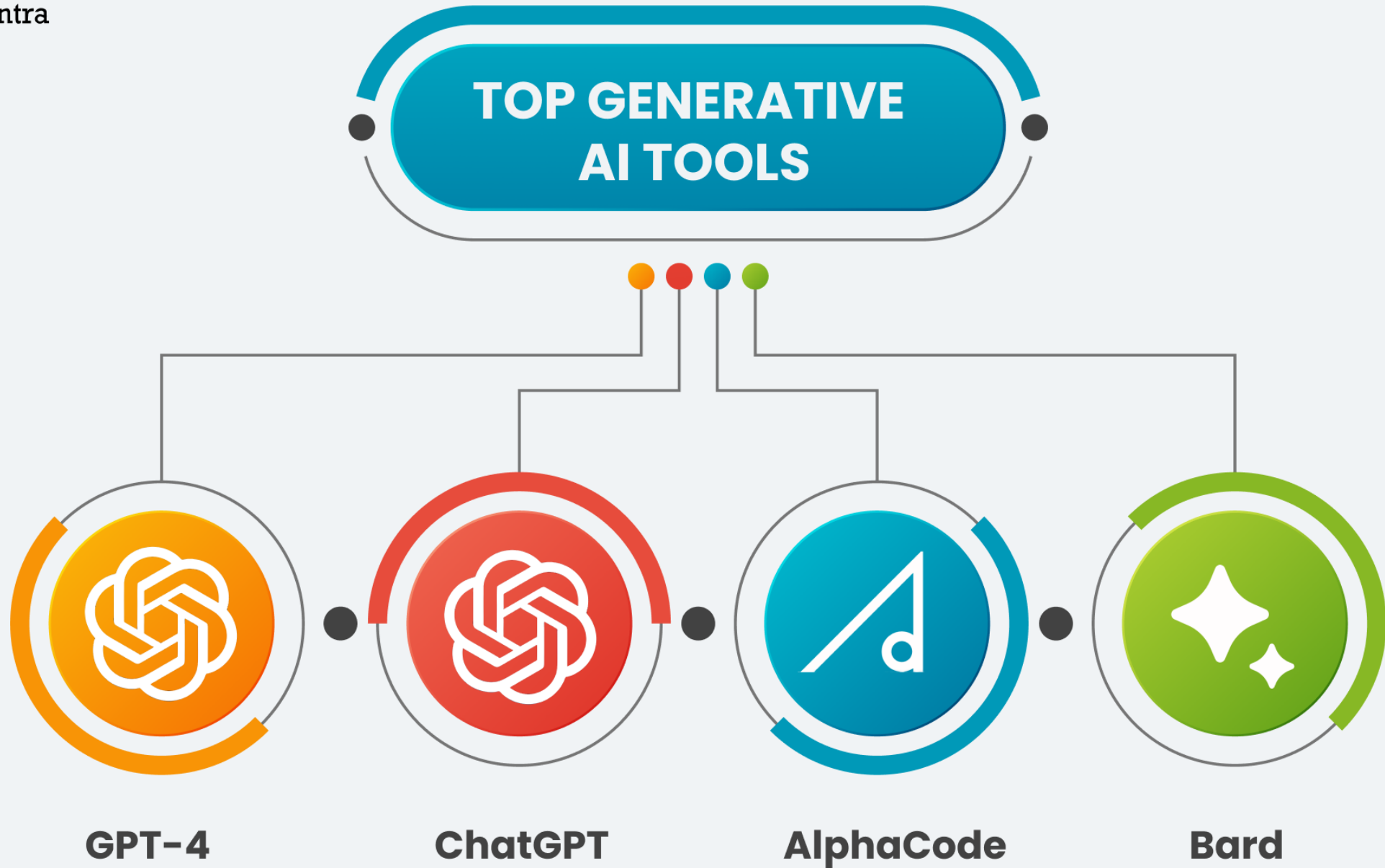
PERFORM
REASONING AND
SUMMARIZATION



ACT AS A “CO-
PILOT” FOR WORK

Top 3 LLMs at a Glance

Model	Developed By	Key Strengths	Unique Features
GPT-4	OpenAI (Microsoft-backed)	Industry-leading accuracy & creativity	Plugins, Code Interpreter, Copilot integration
Gemini	Google (DeepMind)	Multimodal (text + image + video)	Deep GCP integration, powerful search context
Claude	Anthropic	Safer, ethical, large context window (100k+)	Friendly tone, long document understanding



ChatGPT vs. Gemini vs. Claude vs Llama



Which AI is better in 2025?



<https://www.youtube.com/watch?v=5KiDabAa9JY>



1. Core Strengths & Use Cases

ChatGPT-4 (GPT-4o / 4.5)

Best overall performer in coding,

creative writing, and

multimodal interactions.

Excelled in calculus

problem-solving (94.7% success).

<https://openai.com/index/introducing-gpt-4-5/>

1. Core Strengths & Use Cases

Google Gemini (2.5 Pro/Flash)

Strong in multimodal context

(text+image+audio/video),

reasoning-heavy tasks, and

cost-effective free usage.

<https://gemini.google.com/app>

1. Core Strengths & Use Cases

Anthropic Claude (4 Sonnet / Opus)

Tops safety, ethical alignment,

long-context reasoning

best accuracy in reading comprehension

without hallucinations.

<https://claude.ai/onboarding?returnTo=%2F%3F>

1. Core Strengths & Use Cases

Meta Llama 4 (Scout & Maverick)

Competitive open-source option,

excels in reasoning and

massive-context tasks,

rivaling GPT-4o in benchmarks.

Meta Llama 4 (Scout & Maverick)

https://www.meta.ai/?utm_source=ai_meta_site&utm_medium=web&utm_content=AI_nav&utm_campaign=06112025_moment

2. Performance Highlights



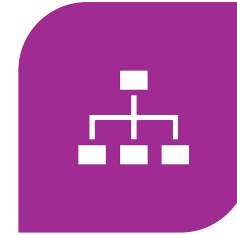
**BENCHMARKS &
ACCURACY**



**CHATGPT-4 LEADS IN
CODING & CALCULUS**



**CLAUDE PRO
FOLLOWS CLOSELY.**



**ON OLYMPICARENA,
GPT-4O AND**



**CLAUDE-3.5-SONNET
TOP GEMINI-1.5-PRO.**

3. Strengths & Weaknesses Table

Feature	ChatGPT-4/4.5	Gemini 2.5	Claude 4	Llama 4
Coding	 Excellent	 Very strong	 Moderate to strong	Good (via Code Llama)
Creative Writing	 Superior prose	 Good (free-tier leader)	 Solid clarity & consistency	Moderate
Reasoning & Math	 Top performance	 Strong	 Excellent accuracy	 Competitive

3. Strengths & Weaknesses Table

Feature	ChatGPT-4/4.5	Gemini 2.5	Claude 4	Llama 4
Multimodality	✓ Text + Images	↑ TOP Natively multimodal	✓ Image + document support	✓ Multimodal support
Safety & Ethics	✓ Good	! Moderate	↑ TOP Rigorous guardrails	Open-source caution

3. Strengths & Weaknesses Table

Feature	ChatGPT-4/4.5	Gemini 2.5	Claude 4	Llama 4
Cost & Access	Paid tiers	Free-tier strong	Free/open beta	Open-source
Long Context Handling	✓ Large	✓ Million-token window	✓ Opus with 200k+	✓ 1M token window

4. Emerging Trends & Developments



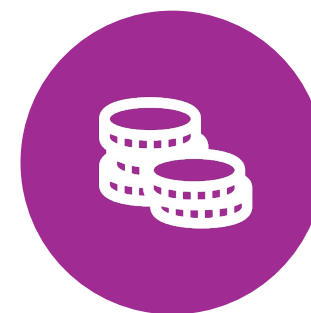
GEMINI 2.5 PRO



INTRODUCES
“DEEP THINK”



REASONING
MODE AND



MILLION-TOKEN
CONTEXTS.

4. Emerging Trends & Developments



**CLAUDE 4
SONNET/OPUS**



ADDS CODE
EXECUTION AND



BROWSING WITH HIGH
SAFETY-FOCUS.

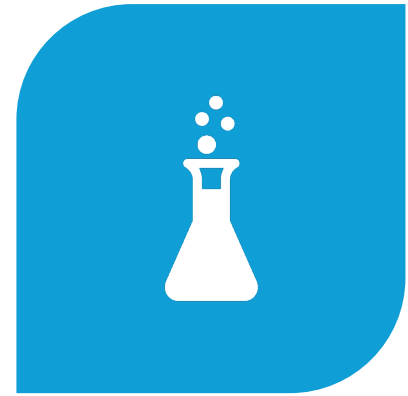
4. Emerging Trends & Developments



LLAMA 4 SCOUT/MAVERICK



ACHIEVE GPT-4 PARITY ON
REASONING BENCHMARKS



USING MIXTURE-OF-
EXPERTS ARCHITECTURE.

2025 Pick by Use Case



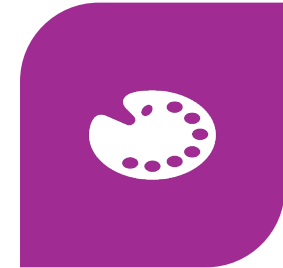
**OVERALL BEST:
CHATGPT-4,**



**WITH
UNMATCHED**



**VERSATILITY
AND**



**CREATIVE
CAPABILITY.**

2025 Pick by Use Case



**BEST FREE AND
MULTIMODAL TOOL**



GEMINI 2.5 FLASH



**STRONG
MULTIMODAL AND**



**GENERAL
PERFORMANCE.**

2025 Pick by Use Case



SAFEST AND MOST
ACCURATE

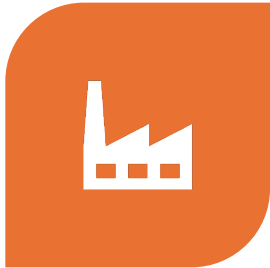


FOR LONG-FORM OR
LEGAL WORK:



CLAUDE 4.

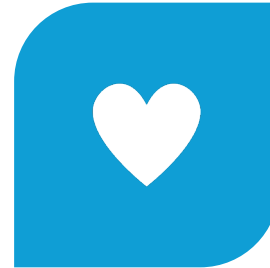
2025 Pick by Use Case



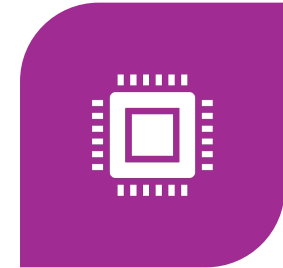
**BEST OPEN-SOURCE
POWERHOUSE**



LLAMA 4 (MAVERICK),

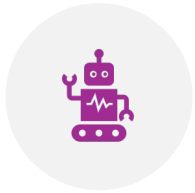


ESPECIALLY FOR



**ON-PREM/SELF-
HOSTED USE.**

Final Take



Each AI is best-in-class for specific goals:



◆ **ChatGPT-4** – elite all-rounder.



■ **Gemini** – multimodal & free utility.



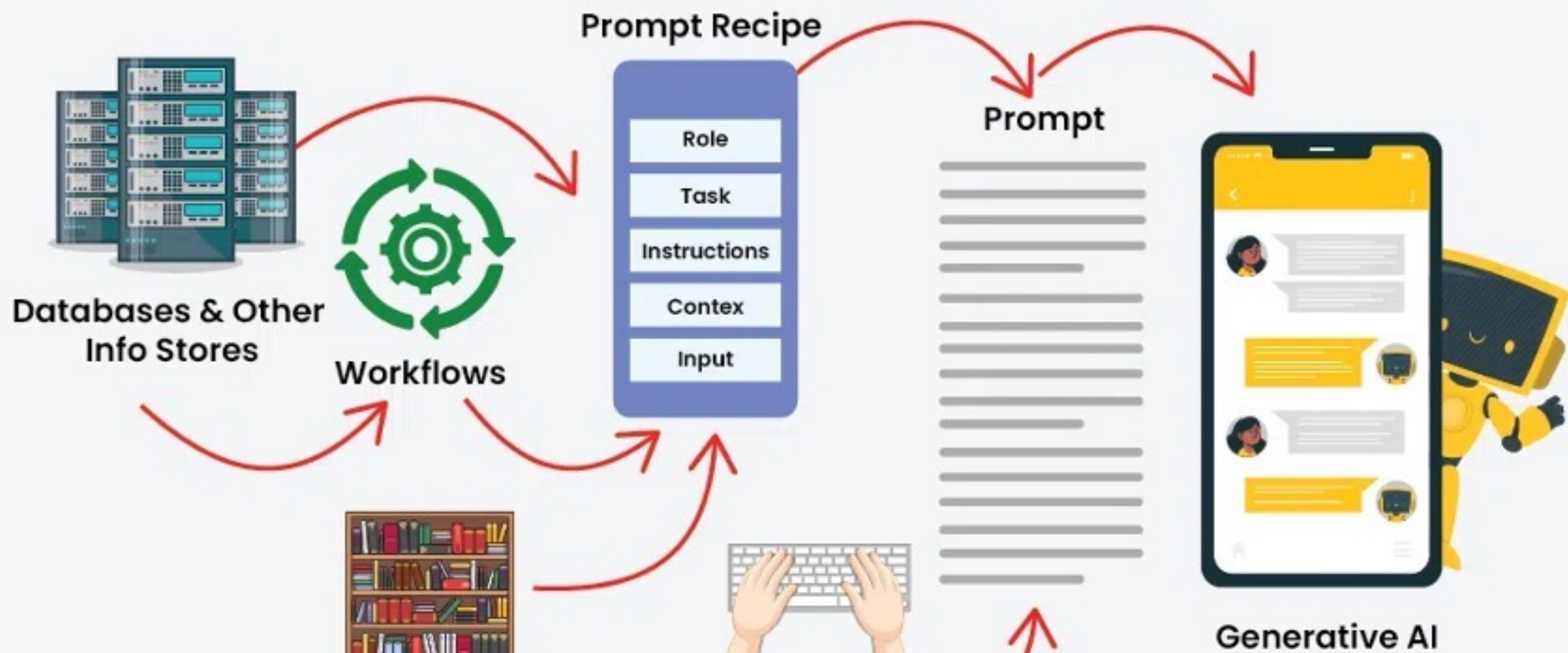
■ **Claude** – safe, ethical, long-context.



■ **Llama** – scalable open-source.



What is an AI Prompt Engineering?



What is Prompt Engineering?



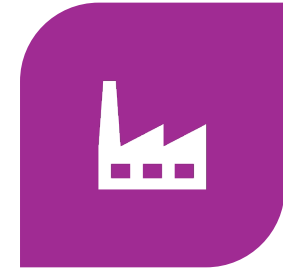
ART AND SCIENCE OF
CRAFTING EFFECTIVE
INPUTS (PROMPTS)



THAT GUIDE AI
MODELS



ESPECIALLY LARGE
LANGUAGE MODELS
(LLMS)



TO PRODUCE
DESIRED OUTPUTS.

What is Prompt Engineering?



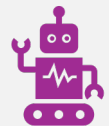
Plays a crucial role in optimizing



how AI systems interpret and



respond to user instructions in



Natural Language Processing (NLP) tasks.

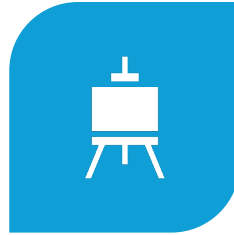
What is Prompt Engineering?



AI PROMPT
ENGINEERING IS



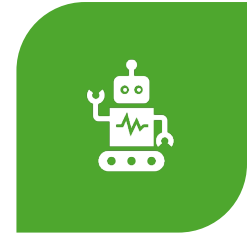
A SPECIALIZED
AREA IN ARTIFICIAL
INTELLIGENCE



FOCUSED ON
DESIGNING,
REFINING, AND



OPTIMIZING
PROMPTS THAT
EFFECTIVELY



COMMUNICATE
USER INTENT TO AI
MODELS.

What is Prompt Engineering?

Prompts act as a bridge between

Human instructions and machine outputs,

allowing models to deliver more accurate,

contextual, and human-like responses.

Why AI Prompt Engineering Matters?



Enhances Model Accuracy:



Well-crafted prompts improve AI output quality.



Tailors Outputs:



Custom prompts align responses to user-specific needs.

Why AI Prompt Engineering Matters?

Reduces Bias:

Ethical prompt design minimizes harmful or biased outputs.

Improves UX:

Better prompts = smoother and more relevant interactions.

Core Activities in Prompt Engineering



Prompt Design:



Creating concise, structured inputs for desired outcomes.



Prompt Optimization:

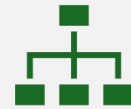


Tuning prompt structure to improve model responses.

Core Activities in Prompt Engineering



Evaluation & Testing:



Assessing effectiveness and refining as needed.



Domain Adaptation:

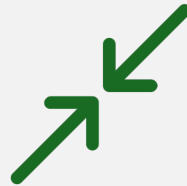


Writing prompts suited for specific industries or tasks.

Core Activities in Prompt Engineering



Ethical Review:



Designing prompts that promote fairness and safety.

Techniques Used



Contextual Prompts:



Add background for relevance.



Specificity:



Reduce vagueness for precise output.

Techniques Used



Iterative Refinement:



Test > Analyze > Modify > Repeat.

Techniques Used



Templates:



Reusable frameworks for consistency.



Experimentation:



Explore open vs. closed-ended prompts.

Best Practices



Understand your AI model.



Keep language clear and concise.



Use examples to guide model behavior.



Avoid ambiguous terms.



Continuously test and improve.

Top Skills Required



Language Proficiency (grammar, syntax)



Creative & Critical Thinking



Data Analysis & Interpretation



Technical Understanding of AI/ML



Attention to Detail

Top Skills Required



Ethical Awareness



Communication & Collaboration



Industry / Domain Knowledge

Trends in the Industry



Rise of Conversational AI & Virtual Assistants



Focus on Personalized Interactions



Integration into domain-specific tools



Cloud-based AI-as-a-Service growth



Safety & alignment in AI responses

Career Paths After Learning Prompt Engineering



AI Prompt Engineer



AI Consultant



AI Product Manager



Prompt Architect

Applications Across Domains



Customer Support:



Virtual assistants & chatbots



Content Creation:



Blogs, social media, articles



Education: Quizzes, flashcards, study guides

Applications Across Domains



Healthcare:



Report generation,
symptom interpretation



Research:



Literature summaries,
trend analysis

Conclusion

Prompt Engineering is a powerful and

emerging field that empowers AI systems

to understand and execute tasks more effectively.

Conclusion



Combining language skills, creativity, ethics, and



technical expertise,



it opens a gateway to advanced



AI-human collaboration and



innovation across industries.

Think of it as

Talking to AI the way

you'd brief your smartest assistant

the better your brief,

the better the result.

Why Prompt Engineering Matters in Automotive

Without Prompt Engineering	With Prompt Engineering
Generic, vague answers	Contextual, technical, precise outputs
Hallucinations or errors	Verified, structured responses with instructions
Missed business value	Maximum ROI from every AI query or automation



Example Applications in Automotive



10 Top Applications of AI in the Automotive Industry



- 1 Advanced Driver Assistance Systems (ADAS)
- 2 Autonomous Vehicles
- 3 Personalized Self-Driving with Smart Assistants
- 4 AI in Automotive Manufacturing
- 5 Voice-Activated Functioning
- 6 Supply Chain Management
- 7 Route Optimization
- 8 AI For Automotive Design
- 9 Predictive Vehicle Maintenance
- 10 Quality Control and Defect Detection

1. Service Manual Generation



 Prompt:



“Write a step-by-step guide on replacing the air filter in a 2024 Toyota Innova Hybrid, suitable for dealership service staff.”

1. Service Manual Generation



Output:

Structured format

Localized to India

Safety precautions + tool list

Visual cues (if model supports images)

2. Driver FAQ Chatbot



 Prompt:



“Create a conversation script for an AI chatbot that answers: ‘Why is my EV charging slowly at home?’ Include possible reasons and recommended actions.”

2. Driver FAQ Chatbot



 Output:



Checks charger type, power supply, battery age



Suggests contacting support or scheduling service



Tone: friendly + technically accurate

3. Marketing Comparison Copy

 Prompt:

Compare the 2025 Tata Nexon EV and MG ZS EV in a customer-friendly tone — highlight performance, range, features, and price.

3. Marketing Comparison Copy



 Output:



Side-by-side comparison



Simplified for buyers



Ideal for brochures or dealer websites

4. Predictive Maintenance Rule Creation



 Prompt:



Based on sensor logs showing



engine temp > 95°C + vibration > 0.4 mm/s



for >10 minutes,



create a rule that flags potential EV motor failure.

4. Predictive Maintenance Rule Creation



OUTPUT:



SQL OR PYTHON
LOGIC



ALERT CONDITION
TEMPLATE



INTEGRATION-
READY SNIPPET



Prompt Engineering Techniques

Technique	Example Prompt Snippet
Be specific	“Generate checklist for EV battery QC at 25°C”
Give context	“Assume you are a Toyota service engineer...”
Ask for format	“Provide output as a JSON list of instructions”
Set role & audience	“Write for a new EV technician with 1 year of experience”
Add constraints	“Limit response to 200 words, and add 3 bullet points”

Prompt Recipe (Framework)

Role:

You are a {service engineer/QA expert/UX designer}

Task:

Help with {writing guide/checking compliance/explaining concept}

Prompt Recipe (Framework)



Context:



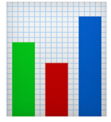
For a {2025 EV model, Indian market, low power charging scenario}



Output Format:



{Checklist, Script, Code, FAQ, Table, Comparison}

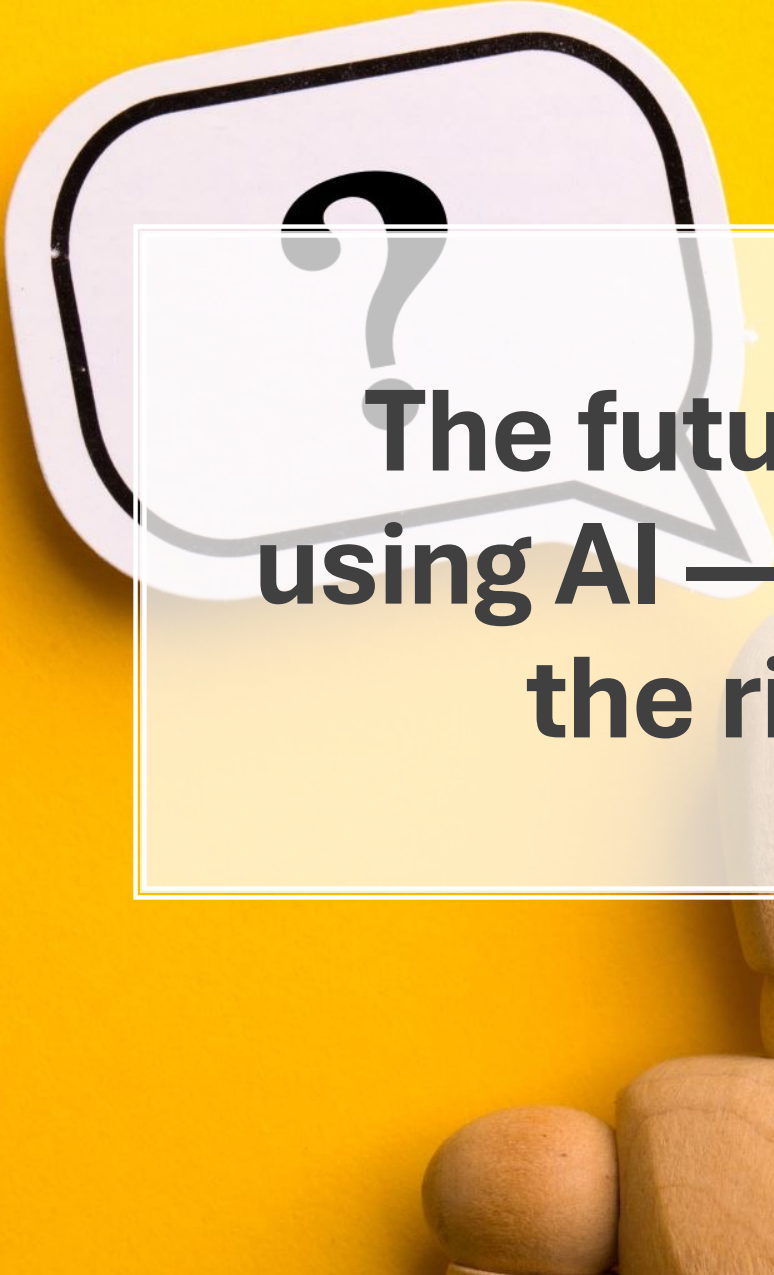


Real-World Impact

Team	Prompt Use Case	Outcome
R&D	Draft failure test cases using design inputs	Faster validation
Customer Service	Generate 100 FAQs for EV issues	Reduced call center load
Compliance	Summarize AIS/ISO standards for engineers	Better audit readiness
HR	Auto-create job descriptions for EV roles	50% faster hiring process

✨ Summary

-  **Prompt Engineering = Productivity Unlock**
- It enables your team to:
- Build smarter AI tools
- Communicate faster
- Automate more processes
- Deliver better decisions

A wooden robot figure is positioned at the bottom of the frame. Its head is a light-colored wooden sphere with a large, dark question mark on it. The robot's body is made of several stacked wooden blocks, and its arms and legs are also made of similar blocks. The background is a solid, bright yellow.

**The future isn't just about
using AI — it's about asking AI
the right questions.**

Use Case

**Auto-Generate Driver Manuals from Technical
Specs**

Objective

Use **Generative AI** (LLMs like GPT, Claude, Gemini)

to automatically convert **technical specification documents**

into **user-friendly driver manuals**,

saving time, ensuring consistency, and

improving customer experience.



What Happens Today (Manual Process)

Step	Challenges
Engineers draft tech specs	Too technical for drivers
Manual writers interpret specs	Time-consuming, error-prone
Language localization	Manual for each language done separately



What Happens with Generative AI

Input	Output
Engineering spec sheet (PDF/Excel/Doc)	Well-structured, simplified driver manual
Format and tone instructions	Brand-consistent, region-specific documents
Language tag	Auto-translated manuals in 10+ languages



Example Input (Spec Snippet)



Battery: Lithium-ion, 40 kWh



Range: 320 km (WLTP)



Charging time: 6.5 hrs (AC), 1 hr (DC fast)



Brake system: Regenerative + Disc brakes



Drive modes: Eco, City, Sport



Example Prompt to LLM

You are an automotive documentation specialist.

Convert the following technical specifications into a friendly, easy-to-understand section for an EV driver manual.

Explain in a tone suitable for non-technical users.

Format as a table + usage tips. Translate to Hindi.

Feature	Details
Battery Type	Lithium-ion, 40 kWh
Range	Up to 320 km (as per WLTP standards)
Charging Time	AC (home charger): 6.5 hrs 

Battery & Charging



Example Output: Auto-Generated Manual Snippet

Battery & Charging

- DC fast charge: 1 hr  |
-  **Tip:** For daily use, charge between 20% and 80% to extend battery life.



Drive Modes



Eco Mode: Maximum efficiency for longer drives



City Mode: Balanced performance for daily traffic



Sport Mode: Enhanced acceleration for highway overtakes

 **Hindi
Translation
Output (if
needed)**



“बैटरी टाइप: लिथियम-आयन, 40
kWh



रेंज: 320 किमी (WLTP के अनुसार)



चार्जिंग समय: 6.5 घंटे (एसी), 1 घंटा
(डीसी फास्ट)”



Benefits to Automotive Companies

Benefit	Value Delivered
 Time savings	70–80% faster than traditional manual writing
 Cost reduction	Fewer writers and translators needed
 Multilingual reach	Auto-translate to 10+ languages (with context)
 Standardization	Consistent tone, style, and branding
 Enhanced customer experience	Manuals that are actually read and used

The background is a vibrant blue digital landscape. On the left, a portion of a globe is visible, showing cloud patterns. The foreground is filled with a grid of binary digits (0s and 1s) that recede into the distance. A bright light source on the right creates a strong lens flare and illuminates the scene. Swirling, ethereal lines of light and more binary code are scattered throughout the composition, giving it a sense of motion and high-tech connectivity.

Real-World Examples



Product Development, Engineering and R&D

Automation, Productivity
Improvement, Process
Optimization



Customer Experience

Interactive In-car driver
assistance
Personalization and
Customization



Connected and Autonomous

Intelligent routing and navigation
Predictive maintenance
Synthetic Data Generation for
ADAS-AD features



Sales and Marketing

Sales and Pricing Optimization



Supply Chain Management

Demand Forecasting and
Inventory Optimization

Hyundai

Used AI to create

interactive driver guides

in regional languages.

Tesla



Auto-generates **in-app help manuals** and



pushes updates based on



vehicle software version.

MG Motors India



USING **GENAI +
SPEECH INTERFACE**



FOR VOICE-
ACCESSIBLE MANUALS



IN **HINDI AND TAMIL.**

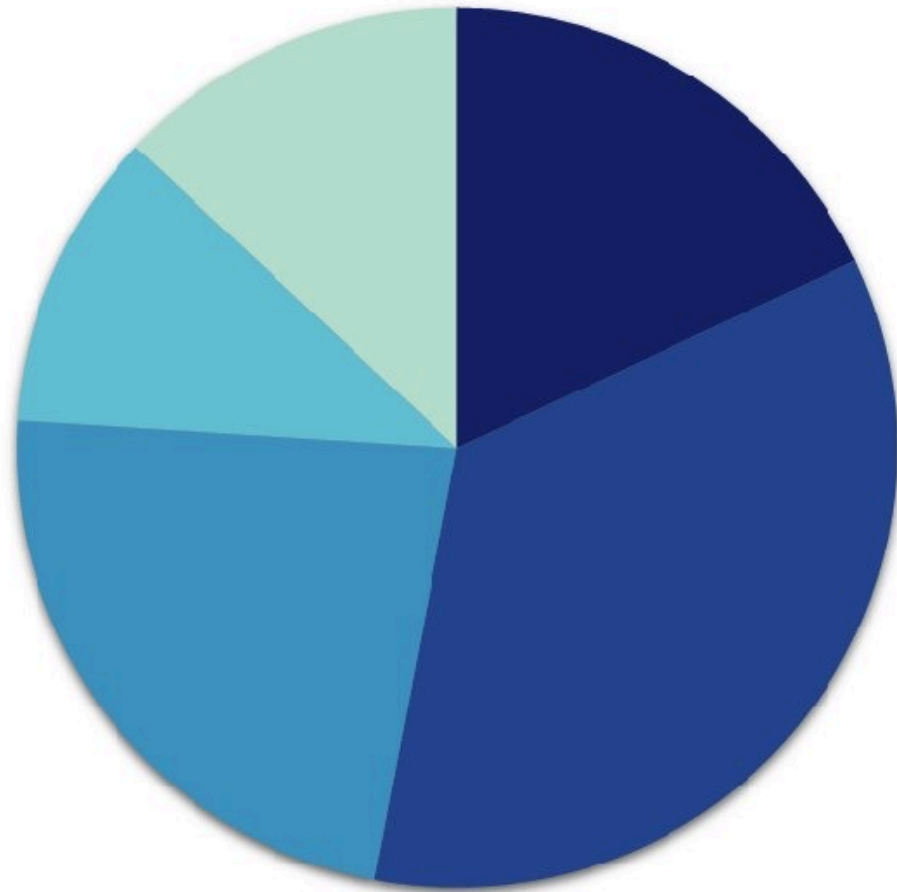


Implementation Workflow

- Upload tech spec documents (PDF, Excel, XML)
- Run prompt template via GPT/Gemini/Claude
- Localize output using multilingual APIs
- Export as PDF, HTML, or embed in vehicle infotainment app
- Auto-update manual based on software or part changes

Generative Ai In Automotive Market

Share, Based on Application, 2023 (%)



- Advanced Driver Assistance Systems (ADAS)
- Connected Car Technologies
- Autonomous Driving Technologies
- Human-Machine Interfaces (HMIs)
- Other Applications



MarketResearchBIZ
WIDE RANGE OF GLOBAL MARKET REPORTS

335.0

**Total Market Size
(USD Million), 2023**

23.4%

**CAGR
2023-2033**





User Readiness for Generative AI in Automotive

75%



of weekly in-car voice AI users are likely to use Generative AI

50%



of weekly drivers are likely to use Generative AI

28%



of drivers who rarely or never use in-car voice AI are likely to use Generative AI



Business Benefits: Gen AI in Automotive

For Businesses	For Consumers
 Efficient production & cost savings	 Personalized in-car experiences
 Adaptive supply chains	 Real-time traffic & safety insights
 AI-powered design & customization	 Predictive maintenance & alerts
 Faster product development	 Personalized recommendations while shopping



Top Automotive Companies Using GenAI

Company	Gen AI Application
Mercedes-Benz	ChatGPT in voice assistants
Toyota	GenAI-aided EV design.
CarMax	GPT-3.5 for automated car descriptions
BMW	AWS-powered GenAI driver assist system
CarGurus	ChatGPT plugin for car search
Master of Code	Multi-dealer conversational AI VA

How generative AI is transforming in-vehicle voice experiences

<https://www.youtube.com/watch?v=xkwwIDgrmbk>



Case Study: Using Generative AI for Multilingual Manual Generation in Automotive

Surendra Panpaliya

Objective



Automate the generation of **driver and service manuals**



in multiple languages using **GenAI**,



to improve documentation speed, accuracy, and



global accessibility especially for diverse markets like



India, Europe, and Southeast Asia.

Client Profile: Global Automotive OEM

Hyundai, Tata Motors, or Volkswagen

Product line:

Electric vehicles (EVs), hybrid SUVs

Client Profile:
Global
Automotive
OEM

Challenge:

Each vehicle variant requires a **50–100 page manual**

Languages needed:

English, Hindi, Tamil, French,
German, Arabic

Client Profile: Global Automotive OEM



Prior process:



Manual translation + localization via agencies



Time per manual: ~4 weeks



Cost: ₹8–10 lakhs per vehicle variant

Problem Statement

High cost and time to generate

**accurate, region-specific
manuals**

Inconsistent tone and formatting

across language versions

Problem Statement



Frequent rework due to



model-specific technical updates



Delay in manual availability affects



vehicle launches and service readiness

GenAI-Based Solution



Tools Used:



OpenAI GPT-4 for content generation



Gemini for translation + layout refinement



Claude for reviewing and summarizing compliance clauses



Integrated with **Google Docs, Canva, and Adobe InDesign**

Process Workflow



Input:



Technical specification sheets
and



engineering design docs

Process Workflow



Prompt:



“You are a documentation expert.



Create a user-friendly EV owner’s manual



for Model Z, in Hindi, with visuals,



safety instructions, and charging FAQs.”

Process Workflow



GenAI Output:



Structured manual in Hindi with simplified tone



Visual aids auto-generated from diagrams



Safety instructions and local regulation references

Process Workflow



Translation Layer:



Auto-translate and refine content in



Tamil, German, and Arabic with **prompt constraints**



like tone, formatting, and glossary control.

Process Workflow



Human-in-the-loop QA:



Technical review and



compliance validation



in target languages.

Example Manual Snippet: Hindi Version

 मॉडल Z ईवी – उपयोगकर्ता गाइड

 बैटरी जानकारी:- बैटरी प्रकार:

लिथियम-आयन 45 kWh-

रेंज: 325 किमी (WLTP)

**Example
Manual
Snippet: Hindi
Version**

⚡ चार्जिंग निर्देश:-

होम चार्जर से: 7 घंटे

- फास्ट चार्जिंग स्टेशन से: 1.2 घंटे

**Example
Manual
Snippet: Hindi
Version**

! सुरक्षा निर्देश

**: - चार्जिंग के समय गीले हाथों से
उपकरण न छुएं**

**- उच्च वोल्टेज सिस्टम को केवल
प्रमाणित तकनीशियन ही छुएं**

Business Impact

Metric	Before (Manual)	After (GenAI-assisted)	% Improvement
Manual creation time	4 weeks	4 days	85% faster
Translation/localization	₹5–7 lakh	₹75K–₹1 lakh	80% cheaper
Errors & revisions	20+ per version	<5 with human QA	75% reduction
Launch readiness	Often delayed	Ready before vehicle ships	On-time



Real-World Application Examples

Automotive Brand	GenAI Application Example
Tata Motors	Auto-generating regional language EV manuals (Hindi, Marathi)
Hyundai	Interactive GenAI-based help inside infotainment system
Renault	GenAI-assisted translation of safety recall documents
Mahindra	ChatGPT-powered voice assistant for vehicle feature guidance



Leadership Takeaways

Decision Factor	GenAI Advantage
Time-to-market	Reduces documentation bottlenecks
Brand experience	Offers regionally tailored manuals
Compliance support	Ensures localization to safety/legal norms
Resource optimization	Reduces dependency on translation vendors
Multichannel support	Manuals available via app, print, and voice

 **Future
Vision**

Voice-activated manuals

using LLMs + TTS in native languages

AR-enabled guides generated from GenAI

(e.g., “how to replace a tire”)

Connected car FAQs via in-vehicle chatbots

linked to dynamic GenAI manuals

Summary



Using GenAI for multilingual manual generation
delivers:



80% faster output



75% cost savings



Higher customer satisfaction



Compliance-ready multilingual documentation



Unified tone across geographies

Happy Learning@!!
Thanks for Your
Patience 😊

Surendra Panpaliya
GKTCS Innovations

